

PENETRATION TESTING REPORT

FOOTPRINTING, RECONNAISSANCE & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name	Malehloa Seroke
Program/Batch	B082-Networkwalks
Date	17 September 2026
Modules Completed	W2-PM1, W2-PM4, W2-PM5, Zenmap Network Scanning
Client/Target	Networkwalks Training Environment, Microsoft.com Training Target, Own Local LAN
Permission Secured	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. LIABILITY DISCLAIMER

This penetration testing report was prepared for educational and training purposes as part of the Networkwalks Cybersecurity Professional B082 program.

All activities were performed against authorized targets or the tester's own local network. No unauthorized access, exploitation, or disruption was performed.

The information presented in this report represents the results observed during the practical exercises and should not be considered a full security assessment of the target environments.

2. INTRODUCTION

The purpose of this project was to perform practical cybersecurity activities covering the reconnaissance, footprinting, and network scanning phases of penetration testing.

The exercises involved gathering publicly available information using Kali Linux tools, performing passive reconnaissance with theHarvester, and identifying live devices on a local network using Zenmap/Nmap.

The activities helped demonstrate how information can be collected before conducting further security testing.

3. TOOLS USED

The following tools were used during the practical exercises:

Tool	Purpose
WHOIS	Domain registration information
WhatWeb	Website technology identification
NSLookup	DNS and IP resolution
CURL	HTTP header information
WAFW00F	WAF detection
DNSRecon	DNS enumeration
theHarvester	Passive reconnaissance
Zenmap/Nmap	Network scanning and host discovery
Kali Linux	Reconnaissance environment
Windows Command Prompt	Local network information

4. ACTIVITIES PERFORMED

4.1 FOOTPRINTING & RECONNAISSANCE

Reconnaissance was performed against networkwalks.com using several Kali Linux tools.

WHOIS was used to collect domain registration information. WhatWeb was used to identify technologies used by the website. NSLookup was used to identify DNS and IP information, while CURL was used to review HTTP response headers.

WAFW00F was used to identify whether a Web Application Firewall could be detected, and DNSRecon was used to gather DNS-related information.

These tools provided different pieces of information that can be combined to build an initial understanding of the target's external footprint.

4.2 PASSIVE RECONNAISSANCE WITH THEHARVESTER

theHarvester was used against microsoft.com as part of the authorised training exercise.

Using the Baidu source with a limit of 1000, theHarvester identified one email address and 12 hosts.

Using all available sources with a limit of 50 produced a larger dataset containing approximately 9,960 hosts/subdomains, 147 IP addresses, 3 email addresses and 9 ASNs.

The results demonstrated how publicly available information can provide an extensive view of an organisation's external digital footprint.

4.3 NETWORK SCANNING WITH ZENMAP

Zenmap/Nmap was used to scan the tester's local network.

The local IPv4 address was identified as 192.168.8.253 with a subnet mask of 255.255.255.0. The local network was therefore identified as 192.168.8.0/24.

A Ping Scan identified **3 live hosts**:

IP Address	MAC Address
192.168.8.1	4C:81:25:1E:D3:48
192.168.8.254	72:D0:59:AC:B6:0A
192.168.8.253	AC-D5-64-AA-AD-31

Zenmap's Topology feature was also used to generate a visual representation of the discovered network.

5. RISK ANALYSIS / IMPACT

The reconnaissance activities demonstrated that publicly available information can reveal domain, DNS, technology, email, host, and infrastructure information.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified technologies used by networkwalks.com	Exposed technology information may assist attackers in identifying technologies that require further security review.	Medium
2	Domain and IP information identifiable	NSLookup resolved networkwalks.com to its associated IP information	Provides information about the network location of the web service and can support further reconnaissance.	Low
3	HTTP technical information exposed	CURL returned HTTP response headers from https://networkwalks.com	HTTP header information may assist with technology fingerprinting and further enumeration.	Low
4	WAF technology identifiable	WAFW00F was used to identify the web application's WAF	Information about the security technology in use may help an attacker understand the target's security architecture.	Low

5	DNS infrastructure information exposed	DNSRecon identified DNS records and related infrastructure information	DNS information can help build a broader profile of the organisation's external infrastructure.	Medium
6	Large number of public hosts and subdomains identified	theHarvester identified approximately 9,960 hosts/subdomains for microsoft.com	A large public footprint provides more information and potential targets for further authorised security assessment.	Medium
7	Public email addresses identified	theHarvester identified 3 email addresses using all available sources	Public email addresses may increase exposure to phishing, spam, and social engineering attempts.	Medium
8	Public IP addresses identified	theHarvester identified 147 IP addresses and 610 host-to-IP entries	Exposed IP information can assist in mapping an organisation's external infrastructure.	Medium
9	Multiple live hosts identified on local network	Zenmap identified 3 live hosts on 192.168.8.0/24	Identifying active devices provides visibility into the local network and highlights the importance of maintaining an authorised device inventory.	Low
10	MAC addresses identifiable on local network	Zenmap identified MAC addresses for the discovered live hosts	MAC address information can assist with identifying and distinguishing devices on the local network.	Low

Risk Key: **Critical**, **Medium**, **Low**

These observations do not by themselves confirm vulnerabilities. Further authorised testing would be required to determine whether any exposed information or systems present an actual security weakness.

6. RECOMMENDATIONS

Based on the observations from the footprinting, passive reconnaissance, and network scanning activities, I recommend the following security improvements:

- **Review publicly exposed technology information**
Organisations should regularly review what information about their web technologies, services, and infrastructure is publicly visible.

- **Keep web technologies updated**
Web platforms, applications, plugins, and other technologies should be regularly updated and reviewed against current security advisories.
- **Review HTTP response headers**
HTTP response headers should be reviewed to identify and reduce unnecessary technical information exposed to external users.
- **Review DNS records regularly**
DNS records should be checked periodically to ensure that only required domains, subdomains, and services are publicly exposed.
- **Review and monitor WAF configuration**
The Web Application Firewall should be properly configured, regularly updated, and monitored to help protect web applications from malicious traffic.
- **Review publicly exposed email addresses**
Organisations should monitor publicly available email addresses and apply appropriate security controls to reduce phishing and social engineering risks.
- **Monitor the external attack surface**
Organisations should regularly review publicly discoverable hosts, subdomains, and IP addresses to identify unnecessary or outdated infrastructure.
- **Perform regular internal network discovery**
Organisations should periodically scan their own networks to identify active devices and maintain visibility of the internal environment.
- **Investigate unknown devices**
Any unexpected device discovered during authorised network scanning should be investigated and verified.
- **Maintain network documentation**
Network topology, IP addresses, devices, and infrastructure information should be documented and updated regularly.
- **Perform security testing with authorization**
Reconnaissance and network scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. CONCLUSION

The practical exercises provided hands-on experience with the reconnaissance and network scanning phases of penetration testing.

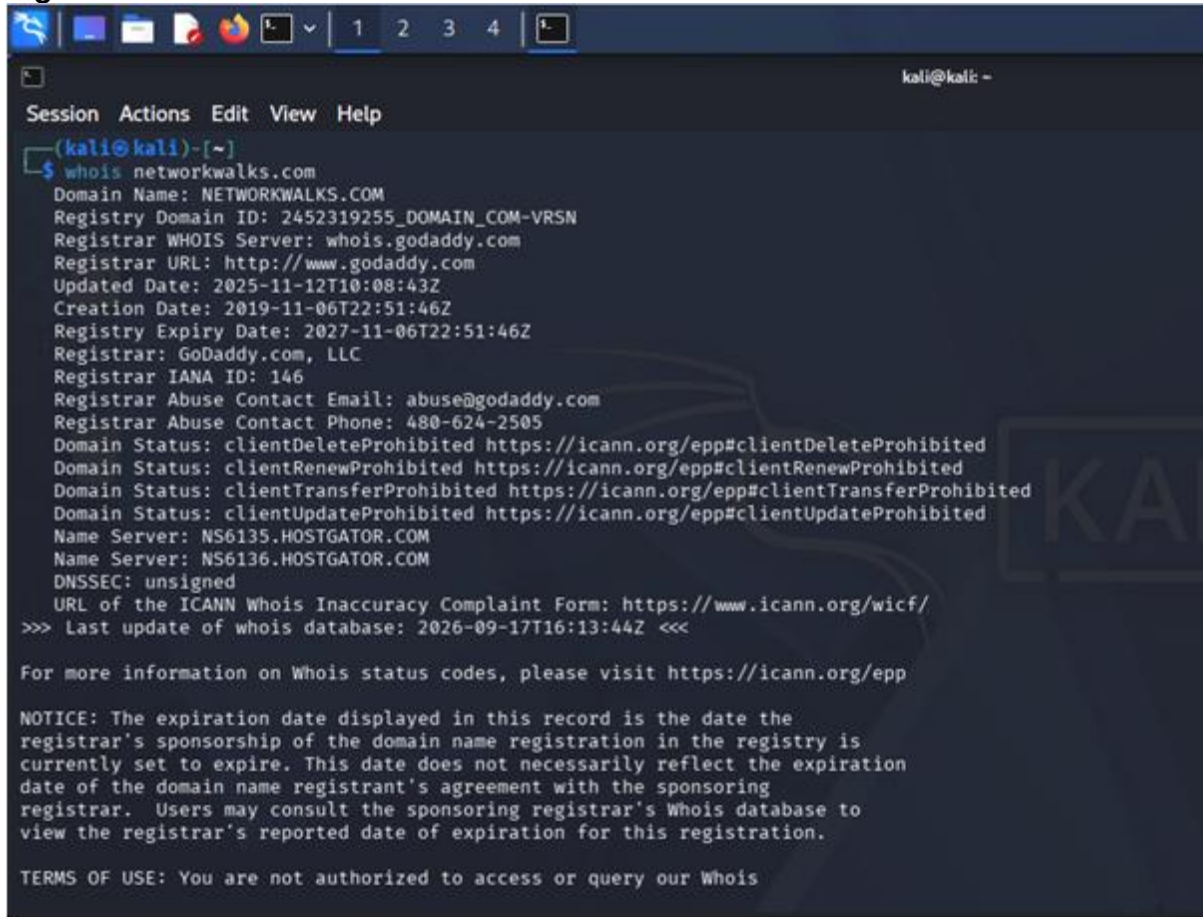
Multiple Kali Linux tools were used to collect information about a target, while theHarvester demonstrated passive reconnaissance using publicly available sources.

Zenmap/Nmap was then used to identify live hosts and network information on the local LAN.

The exercises improved practical understanding of how reconnaissance information is collected, analysed, documented, and used during the early stages of a penetration test.

8. EVIDENCES COLLECTED

Figure 1: WHOIS results for networkwalks.com



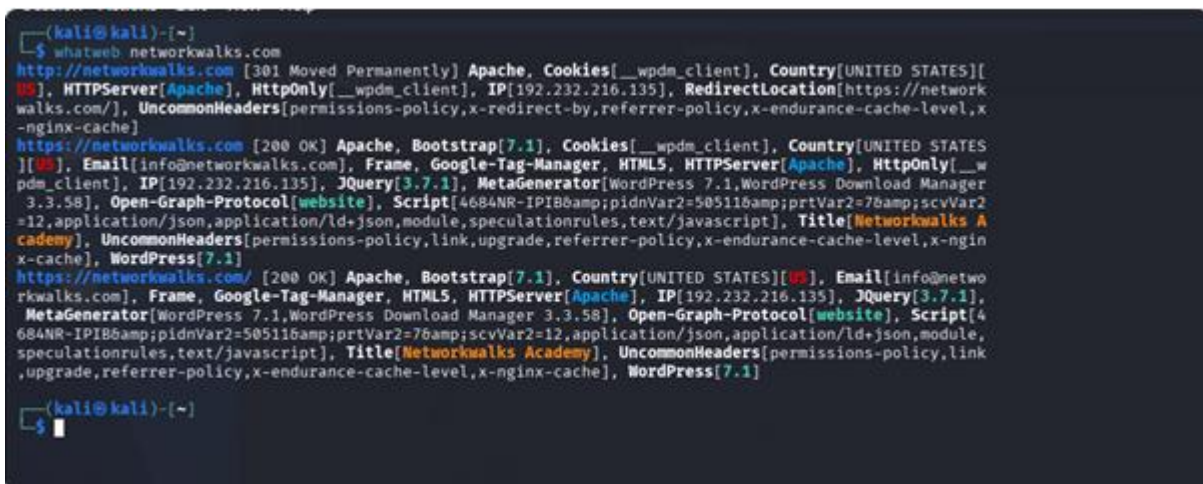
```
(kali@kali)~$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-17T16:13:44Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
```

Figure 2: WhatWeb results for networkwalks.com



```
(kali@kali)~$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wvdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wvdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[__wvdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wvdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IP1B8amp;pidnVar2=505116amp;pvtVar2=76amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IP1B8amp;pidnVar2=505116amp;pvtVar2=76amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```

```
(kali㉿kali)-[~]  
$ nslookup networkwalks.com  
Server:      8.8.8.8  
Address:     8.8.8.8#53  
  
Non-authoritative answer:  
Name:   networkwalks.com  
Address: 192.232.216.135
```

```
kali@kali:~$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/?; rel="https://api.w.org/">, <https://networkwalks.com/wp-json/wp/v2/pages/53; rel="alternate">; title="JSON"; type="application/json">, <https://networkwalks.com/?; rel=shortlink
set-cookie: wpcom_client=56b8ebca274c2b7700ba2ebd1d7b4e; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Thu, 17 Sep 2026 17:01:49 GMT
server: Apache
```

```

(kali㉿kali)-[~]
$ wafw00f networkwalks.com

    ( W00f! )

Trash

~ WAFW00F : v2.4.2 ~

The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

(kali㉿kali)-[~]
$
  
```

404 Hack Not Found

405 Not Allowed

403 Forbidden

502 Bad Gateway

500 Internal Error

```
(kali@kali)-[~]
└─$ dnsrecon -d networkwalks.com
2026-09-17T13:04:21.090345-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T13:04:21.091879-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T13:04:21.952779-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-17T13:04:22.590486-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-17T13:04:23.025450-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-17T13:04:23.681305-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-17T13:04:23.681677-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-17T13:04:24.317189-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-17T13:04:25.305838-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-17T13:04:25.637636-0400 INFO A networkwalks.com 192.232.216.135
2026-09-17T13:04:26.687687-0400 INFO TXT networkwalks.com google-site-verification=rr04ErMqHoWY3XemniZDNVK4q75X-Ij-mjgEg-UsYi
2026-09-17T13:04:26.687989-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com -all
2026-09-17T13:04:27.474844-0400 INFO Enumerating SRV Records
2026-09-17T13:04:31.243361-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-09-17T13:04:31.243555-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-09-17T13:04:31.243647-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-09-17T13:04:31.243723-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-09-17T13:04:31.243793-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-09-17T13:04:31.243859-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-09-17T13:04:31.243924-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-09-17T13:04:31.243987-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-09-17T13:04:31.246408-0400 INFO 8 Records Found
2026-09-17T13:04:31.246652-0400 INFO Completed enumeration for domain: networkwalks.com

(kali@kali)-[~]
└─$
```

```
(kali@kali)-[~]
$ theHarvester -d microsoft.com -l 1000 -b baidu
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
 *                                     *
 * | _ | _ |   ^   ^   | _ | _ |   | *
 * | | | _ | _ |   v   v   | | | _ | *
 * | | | | | | |   C   I   | | | | | *
 * | _ | _ |   v   v   | _ | _ |   | *
 *                                     *
 * theHarvester 4.10.1                *
 * Coded by Christian Martorella      *
 * Edge-Security Research             *
 * cmartorella@edge-security.com     *
 *                                    *
 *****

[*] Target: microsoft.com

[*] Searching Baidu.

[*] No IPs found.

[*] Emails found: 1
-----
join-ms@microsoft.com

[*] No people found.

[*] Hosts found: 12
-----
account.microsoft.com
accountprotection.microsoft.com
```

Figure 8: theHarvester all-source results



```
Session Actions Edit View Help
zoomeye

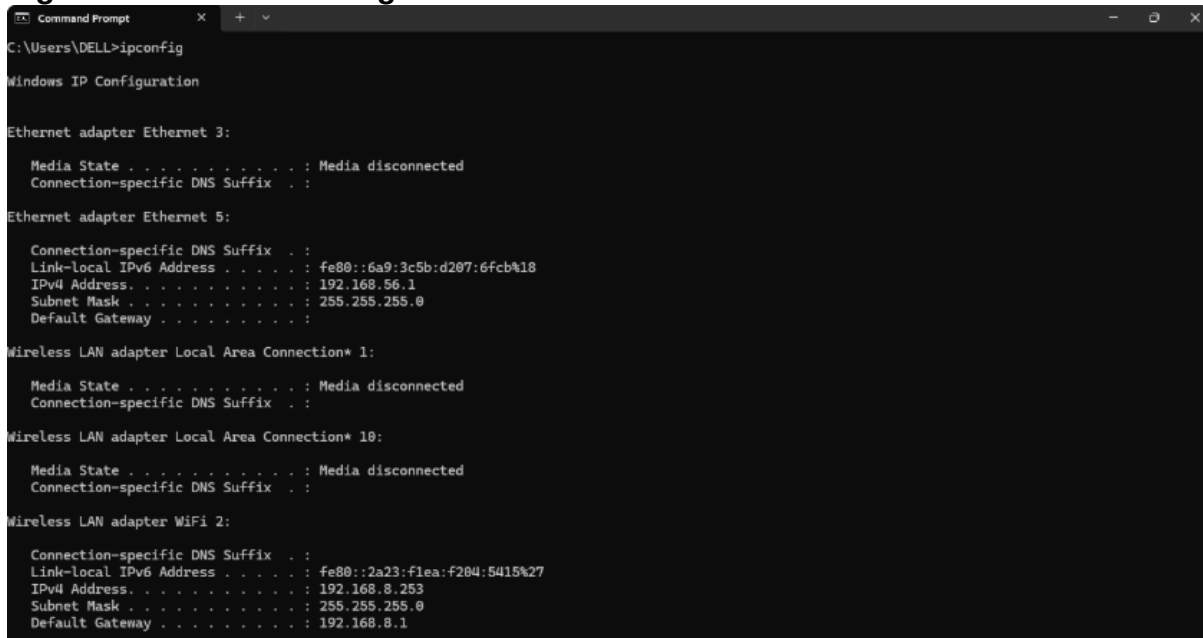
kali@kali:~$ theHarvester -d microsoft.com -l 50 -b all
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*****

[*] Target: microsoft.com

Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process Bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml

[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform BuiltWith search for word: 'microsoft.com'
```

Figure 10: Local IP configuration



```
Command Prompt
C:\Users\DELL>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 5:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::6a9:3c5b:d207:6fcb%18
    IPv4 Address. . . . . : 192.168.86.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::2a23:f1ea:f204:5415%27
    IPv4 Address. . . . . : 192.168.8.253
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.8.1
```

Figure 11: Zenmap Ping Scan and Live Hosts

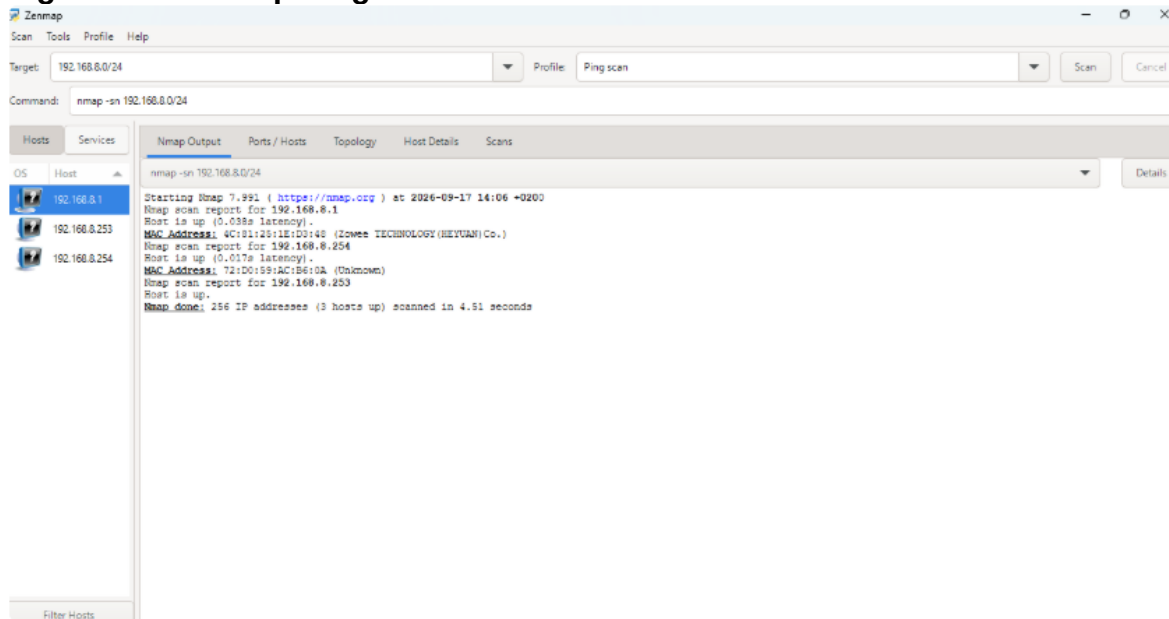
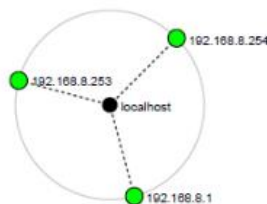


Figure 14: Zenmap network topology



AUTHOR AND PROJECT INFORMATION

Pentester: Malehloa Seroke

Program: Cybersecurity Professional B082 Intership

Training Provider: Networkwalks

Project: Week 2 – Footprinting, Reconnaissance & Network Scanning

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